



8655 - Dynamic modeling of the shocking dust physics in WR 140

Cycle: 4, Proposal Category: AR

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Dr. Christopher Michael Post Russell (PI)	University of Delaware
Dr. Ryan M Lau (CoI)	NOIRLab - (AZ)
Prof. Takashi Onaka (CoI)	University of Tokyo, Graduate School of Science
Hideo Matsuhara (CoI)	Institute of Space & Astronautical Science
Dr. Astrid Lamberts (CoI) (ESA Member)	CNRS, Observatoire de la Cote d'Azur
Dr. Anthony Soulain (CoI) (ESA Member)	Institut de Planetologie et d'Astrophysique de Grenoble
Dr. Diego Calderon (CoI) (ESA Member)	Max Planck Institute for Astrophysics
Dr. Matthew J Hankins (CoI)	Arkansas Tech University
Dr. Thomas I. Madura (CoI)	San Jose State Univ. Research Foundation
Dr. Theodore Raymond Gull (CoI)	NASA Goddard Space Flight Center
Dr. Henrik Hartman (CoI) (ESA Member)	Malmo University
Dr. Noel D. Richardson (CoI)	Embry-Riddle Aeronautical University
Prof. Anthony F.J. Moffat (CoI) (CSA Member)	Universite de Montreal
Dr. Peredur Williams (CoI) (ESA Member)	University of Edinburgh, Institute for Astronomy
Dr. Michael F. Corcoran (CoI)	Catholic University of America
Dr. Kenji Hamaguchi (CoI)	University of Maryland Baltimore County

OBSERVATIONS

ABSTRACT

Carbon-rich Wolf-Rayet star (WC) plus O-star binaries are copious dust producers, with the binary WR 140 being the canonical example of episodic dust-generators. Early Release Science observations using JWST/MIRI/MRS of WR 140 have provided unprecedented spatial and spectral

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resolution of not only the dust structure surrounding the binary, but also the gaseous wind structures whose collisions form the dust. This system is therefore ripe for a dynamic modeling effort to use both the dust and gas diagnostics to constrain how the collision of the stellar winds produces dust. We will undertake hydrodynamic simulations of the colliding winds of WR 140 and then produce radiative transfer calculations on the output in order to directly compare the models with the MRS observables, all while iterating on the model parameters in order to improve the model-to-observation agreement. We will also synthesize the X-ray emission from the hydrodynamic simulations to compare against the many observations with NASA's X-ray fleet, thereby constraining WR 140's full shock structure -- from the apex of the wind-wind collision region directly between the stars to the much larger dust-enriched region -- simultaneously. The end result will be improved constraints on how this important class of objects generates dust, such as grain size, nucleation radius, temperature, and composition.

OBSERVING DESCRIPTION